STAT 405 - Project Report

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GitHub and Contributions

GitHub Repo: https://github.com/isobela/Stat-405-Project

This project was completed collaboratively. Isobel was responsible for implementing and analyzing the Laplace approximation, including model formulation and interpretation of results. Danial implemented the Hamiltonian Monte Carlo (HMC) method in Stan and carried out the corresponding posterior inference. Both authors contributed to writing, editing, and finalizing the report. To ensure equal contribution, both authors reviewed and revised all sections of the report.

Data

We use publicly available crime data from the Vancouver Police Department, containing yearly counts of multiple crime categories across neighbourhoods in Vancouver.

Since the data were provided in PDF format, we manually constructed a dataset in R. Each observation y_{ij} represents the count of crime type j in neighbourhood i.

Data was created as an R data frame object see Appendix A.

```
##               neighbourhood sex_offences assaults robbery break_enter theft_mv
## 1             Arbutus Ridge            0       18       1          25        7
## 2 Central Business District          151     1591     218         438       74
## 3         Dunbar-Southlands            8       22       0          24        7
## 4                  Fairview           16      115       9         164       16
## 5        Grandview-Woodland           27      254      22         121       64
## 6          Hastings-Sunrise           29       97       7          84       31
##   theft_auto theft_under_5k arson mischief offensive_weapons
## 1         41             69     1       34                 4
## 2       1855           4458    68     1864               708
## 3         72             70     1       33                 4
## 4        219            706     6      209                28
## 5        219            507    15      281                44
## 6        222            312     9      140                 9
```

Literature Review

Bayesian methods are widely used for modeling count data, particularly through Poisson likelihoods with conjugate Gamma priors, which provide a natural framework for estimating underlying rates while accounting for uncertainty [1].

Hierarchical Bayesian models extend this approach by introducing shared structure across groups, allowing for partial pooling of information. This has been shown to produce more stable and reliable estimates, especially in settings with sparse or highly variable data [1].

Such approaches have also been applied in crime data analysis, where counts of incidents are modeled across regions or categories to identify underlying patterns while reducing sensitivity to noise in low-count areas. Similar hierarchical Poisson models are commonly used in modeling grouped or spatial count data.

In contrast to existing work, which typically focuses on model development or application alone, our analysis focuses on Vancouver neighbourhood-level crime data and compares two posterior inference methods, Laplace approximation and Hamiltonian Monte Carlo, to evaluate their effectiveness within the same hierarchical model.

Models

We model crime counts y_{ij} using Poisson distributions.

Model 1 (Independent model):

$$y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(\lambda_{ij}), \quad \lambda_{ij} \sim \text{Gamma}(\alpha, \beta)$$

Model 2 (Hierarchical model):

$$y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(\lambda_{ij}), \quad \log(\lambda_{ij}) = \mu + \alpha_i + \beta_j$$

$$\alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2), \quad \beta_j \sim \mathcal{N}(0, \sigma_\beta^2)$$

We impose $\sum_i \alpha_i = 0$ and $\sum_j \beta_j = 0$ for identifiability.

We interpret λ_{ij} as the expected number of crimes per year for crime type j in neighbourhood i, representing a yearly rate without explicit adjustment for exposure or opportunity.

The Poisson distribution is used since the data consist of non-negative integer counts of crimes. This makes it a natural choice for modeling the variability in the number of incidents.

We model $\log(\lambda_{ij}) = \mu + \alpha_i + \beta_j$ so that λ_{ij} remains positive, since crime rates cannot be negative. The log transformation also allows the effects of neighbourhoods and crime types to combine additively in a simple and interpretable way.

We place normal priors on α_i and β_j since these effects can reasonably take both positive and negative values, representing increases or decreases relative to the overall mean. Centering these priors at zero reflects the assumption that, prior to observing the data, no neighbourhood or crime type is expected to systematically deviate from the global average. These priors also introduce regularization and allow for partial pooling, which stabilizes estimates across neighbourhoods and crime types, particularly in cases with limited or highly variable counts.

Questions

Using the `crime_data` and specified models above we hope to answer the following questions using bayesian analysis:

- What are the underlying rates of different categories of crime across Vancouver neighbourhoods?
- What future crime counts might be predicted for selected neighbourhood and crime-type combinations?

Posterior Laplace Approximation

For Model 1, the posterior distribution can be computed exactly because the Gamma prior is conjugate to the Poisson likelihood. Posterior summaries were computed analytically (see Appendix B).

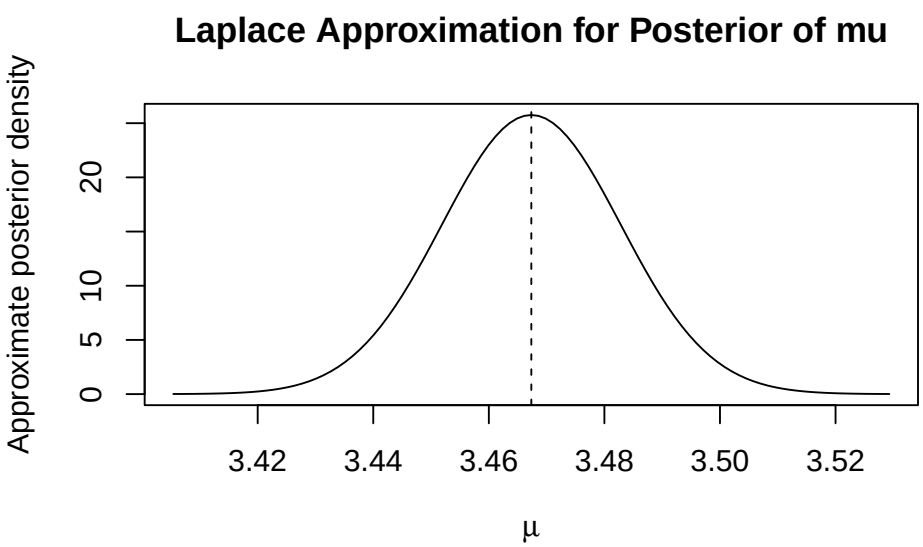
```
##               neighbourhood   crime_type observed_count post_shape post_rate
## 1             Arbutus Ridge sex_offences              0          1         2
## 2 Central Business District sex_offences            151        152         2
## 3         Dunbar-Southlands sex_offences              8          9         2
## 4                  Fairview sex_offences             16         17         2
## 5        Grandview-Woodland sex_offences             27         28         2
## 6          Hastings-Sunrise sex_offences             29         30         2
##   posterior_mean posterior_variance   ci_lower  ci_upper
## 1            0.5               0.25  0.0126589  1.844440
## 2           76.0              38.00 64.3982873 88.548417
## 3            4.5               2.25  2.0576865  7.881595
## 4            8.5               4.25  4.9515632 12.991499
## 5           14.0               7.00  9.3028983 19.641791
## 6           15.0               7.50 10.1204370 20.824419
```

In contrast, for Model 2 the posterior does not have a closed-form expression due to the hierarchical log-linear structure. Therefore, we use a Laplace approximation to obtain an approximate posterior distribution.

We compute the MAP estimate $\hat{\theta}$ numerically and approximate the posterior as

$$\theta \mid y \approx \mathcal{N}\Big(\hat{\theta},\, (-H(\hat{\theta}))^{-1}\Big)\,,$$

where $H(\hat{\theta})$ is the Hessian of the log-posterior (see Appendix C).



The Laplace approximation is most accurate when the posterior is approximately Gaussian near its mode.

Posterior Results and Interpretation

We now compare the posterior results from Model 1 and Model 2 in order to answer our research questions.

Under Model 1, each (i, j) pair is treated independently, and the posterior distributions for λ_{ij} are available in closed form. The posterior means vary widely across neighbourhoods and crime types, reflecting the observed counts. For example, low-count categories (e.g., $y_{ij} = 0$) produce small but nonzero posterior means due to the influence of the prior, while high-count areas such as the Central Business District yield large posterior estimates. However, because Model 1 does not share information across groups, these estimates can be highly variable, particularly for rare events.

In Model 2, we introduce a hierarchical structure

$$\log(\lambda_{ij}) = \mu + \alpha_i + \beta_j,$$

and approximate the posterior using a Laplace approximation. The parameter μ represents the overall average log-crime intensity across all neighbourhoods and crime types. From our results, we obtain $\hat{\mu} \approx 3.47$, corresponding to an average crime rate of

$$\exp(3.47) \approx 32$$

incidents per year. This can be interpreted as a baseline level of crime, with α_i and β_j capturing deviations for specific neighbourhoods and crime categories.

Unlike Model 1, Model 2 allows for partial pooling, meaning that estimates are shrunk toward this overall mean. As a result, extreme values are moderated, and estimates for low-count categories become more stable. This produces more reliable estimates of the underlying crime rates across Vancouver.

To answer our first question, we find that crime rates differ substantially across both neighbourhoods and crime types. High-density areas such as the Central Business District exhibit consistently higher rates across multiple categories, while quieter neighbourhoods show lower overall rates. Model 2 provides a clearer picture of these differences by separating overall trends (μ) from neighbourhood and crime-specific effects.

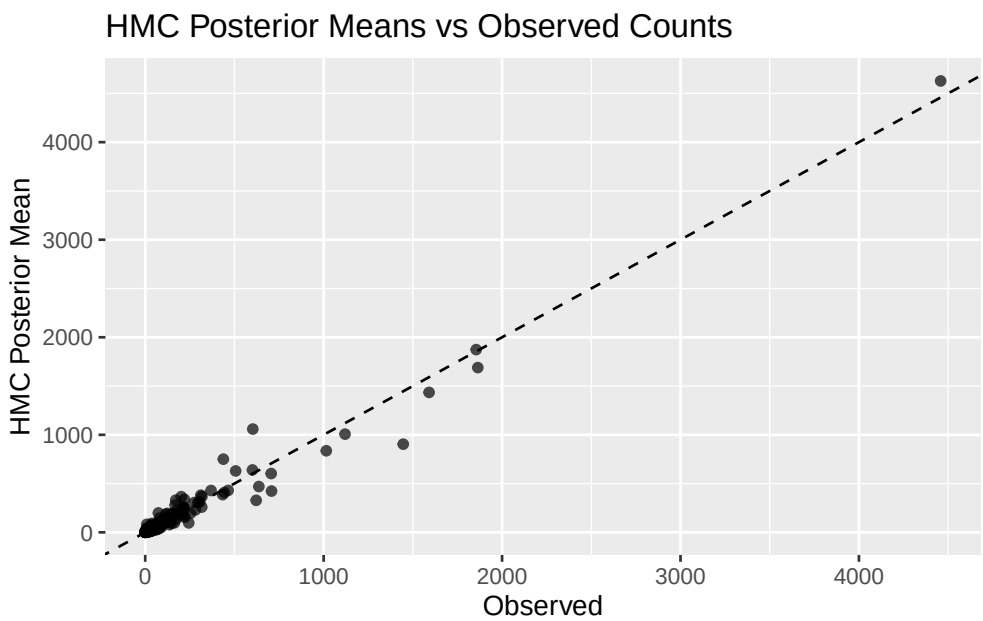
To answer our second question, we use the posterior distributions to predict future crime counts. Under both models, the expected future count for a given (i, j) is approximately equal to the posterior mean of λ_{ij}. This implies that areas with historically high crime counts are expected to remain high, while areas with low counts remain low, though with greater uncertainty. Model 2 improves these predictions by incorporating shared structure, leading to more stable and realistic forecasts, particularly in cases with limited data.

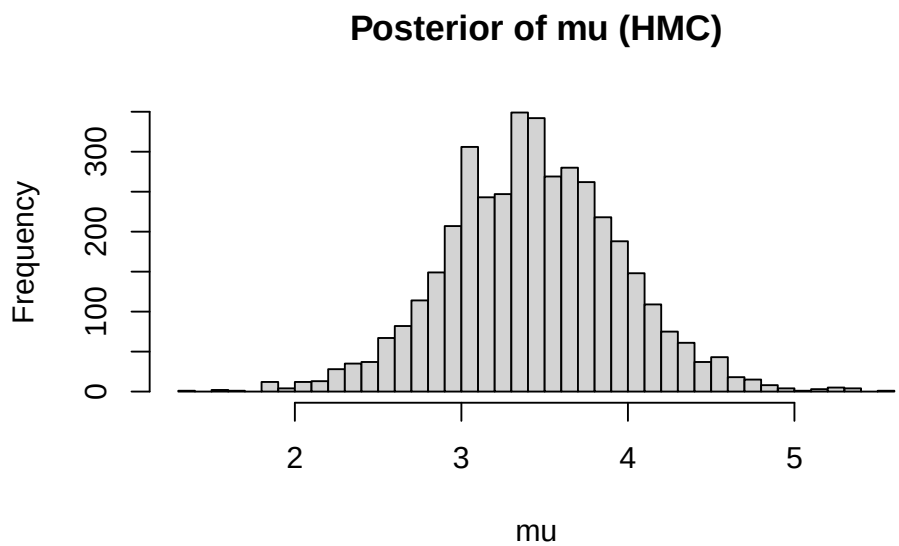
Overall, Model 1 provides a simple baseline with exact posterior computation, while Model 2 offers a more flexible and interpretable framework that captures shared patterns across neighbourhoods and crime types. The Laplace approximation allows us to efficiently perform inference in this more complex model while still obtaining meaningful insights about crime patterns and future risk.

Posterior Inference using HMC

We also perform posterior inference using Hamiltonian Monte Carlo (HMC), implemented via Stan. HMC uses gradient information of the log-posterior to efficiently explore high-dimensional parameter spaces. Implementation details are provided in Appendix D.

```
##                    mean      lower      upper
## lambda[1,1]    3.205872   2.714392   3.748414
## lambda[2,1]  182.518708 167.312312 198.577564
## lambda[3,1]    3.865980   3.300408   4.509737
## lambda[4,1]   23.782742  21.496185  26.134638
## lambda[5,1]   24.828853  22.457833  27.337133
## lambda[6,1]   15.011759  13.470296  16.640574
```





The HMC posterior for μ is approximately symmetric and centered around 3.4–3.5, which closely matches the Laplace approximation. This indicates that the posterior distribution is approximately symmetric and unimodal, making the Gaussian approximation used in the Laplace method reasonable in this setting.

To assess convergence, we examined $\hat{R}$, effective sample size (ESS), and trace plots (see Appendix D). The $\hat{R}$ values were close to 1 and ESS values were sufficiently large, indicating good mixing across chains. The trace plots also show stable behaviour with no signs of non-convergence.

Using the HMC samples, we obtain posterior estimates of λ_{ij} that reflect the underlying crime patterns across neighbourhoods and crime types. In particular, higher predicted values are observed in dense areas such as the Central Business District, while lower values are seen in quieter neighbourhoods, confirming that crime rates vary substantially across both dimensions.

Compared to Model 1, which treats each (i, j) independently, the hierarchical model estimated via HMC produces more stable and moderated estimates due to partial pooling. This reduces variability for low-count categories and prevents extreme estimates driven by sparse data.

The posterior predictive means also align closely with the observed counts, as seen in the scatter plot, indicating that the hierarchical model provides a good fit to the data.

To answer our research questions, the HMC results show that underlying crime rates differ significantly across neighbourhoods and categories, and that future crime counts can be predicted using the posterior mean of λ_{ij}. These predictions suggest that areas with historically high crime counts are expected to remain relatively high, while low-count areas remain lower, with uncertainty captured by the posterior distribution.

Overall, both inference methods produce consistent results, particularly for the global parameter μ, suggesting that the posterior is close to Gaussian in this setting. However, the two approaches differ conceptually: the Laplace approximation relies on a local quadratic approximation around the mode, while HMC explores the full posterior using gradient-based sampling. As a result, HMC is generally more robust, especially in higher-dimensional settings, whereas the Laplace approximation is computationally more efficient.

Discussion and Limitations

In this project, we modeled crime counts across Vancouver neighbourhoods and crime categories using both an independent Poisson-Gamma model and a hierarchical Poisson model. The hierarchical model was more useful overall, as it allows information to be shared across neighbourhoods and crime types, leading to more stable estimates compared to treating each observation independently.

For posterior inference, we used both the Laplace approximation and Hamiltonian Monte Carlo (HMC). The posterior summaries from both methods were very similar, particularly for the parameter μ, suggesting that the Laplace approximation captures the main features of the posterior distribution in this setting. However, HMC remains more reliable in general, as it samples from the full posterior without relying on a Gaussian approximation.

One important limitation is how we defined λ_{ij}. In our setup, it represents the expected number of reported crimes per year, since we only have yearly aggregated data. This means we do not adjust for population or other measures of exposure, so comparisons across neighbourhoods reflect total crime volume rather than per-capita risk. As a result, areas with larger populations or higher activity levels may naturally have higher counts, which the model does not explicitly account for.

Another limitation is that the data only include reported crimes. The true crime rates may differ due to underreporting or variation in reporting behaviour across neighbourhoods, which could introduce bias into our estimates.

Finally, the Laplace approximation depends on the assumption that the posterior appears approximately Gaussian around the mode. While this assumption appears reasonable in our case, it may not hold in more complex models or datasets. In such cases, sampling-based methods like HMC would be more appropriate.

Overall, the hierarchical model combined with Bayesian inference provides a flexible and interpretable framework for understanding crime patterns and making predictions, despite the limitations in the available data.

Appendix

A. Data Construction

```
##                neighbourhood sex_offences assaults robbery break_enter theft_mv
## 1              Arbutus Ridge            0       18       1          25        7
## 2 Central Business District          151     1591     218         438       74
## 3          Dunbar-Southlands            8       22       0          24        7
## 4                   Fairview           16      115       9         164       16
## 5         Grandview-Woodland           27      254      22         121       64
## 6           Hastings-Sunrise           29       97       7          84       31
##   theft_auto theft_under_5k arson mischief offensive_weapons
## 1         41             69     1       34                 4
## 2       1855           4458    68     1864               708
## 3         72             70     1       33                 4
## 4        219            706     6      209                28
## 5        219            507    15      281                44
## 6        222            312     9      140                 9
```

B. Model 1 Posterior Computation

```r
alpha <- 1
beta <- 1
Y <- as.matrix(crime_data[, -1])
post_shape <- alpha + Y
post_rate  <- beta + 1
post_mean <- post_shape / post_rate
post_var  <- post_shape / (post_rate^2)

# 95% credible intervals
post_lower <- qgamma(0.025, shape = post_shape, rate = post_rate)
post_upper <- qgamma(0.975, shape = post_shape, rate = post_rate)
model1_results <- data.frame(
  neighbourhood = rep(crime_data$neighbourhood, times = ncol(Y)),
  crime_type = rep(colnames(Y), each = nrow(Y)),
  observed_count = as.vector(Y),
  post_shape = as.vector(post_shape),
  post_rate = as.vector(post_rate),
  posterior_mean = as.vector(post_mean),
  posterior_variance = as.vector(post_var),
  ci_lower = as.vector(post_lower),
  ci_upper = as.vector(post_upper)
)
```

C. Laplace Approximation Implementation

```r
p <- 1 + (I - 1) + (J - 1)
init <- rep(0, p)
opt <- optim(par = init,
  fn = log_posterior,
  Y = Y,
  sigma_alpha = 1,
  sigma_beta = 1,
```

```r
  method = "BFGS",
  control = list(fnscale = -1, maxit = 50000, reltol = 1e-10),
  hessian = TRUE)
theta_hat <- opt$par
H_hat <- opt$hessian
Sigma_hat <- solve(-H_hat)
se_hat <- sqrt(diag(Sigma_hat))
```

D. HMC Implementation and Diagnostics

```r
stan_code <- "
data {
  int<lower=1> I;
  int<lower=1> J;
  array[I,J] int<lower=0> y;
}
parameters {
  real mu;
  vector[I] alpha_raw;
  vector[J] beta_raw;
  real<lower=0> sigma_alpha;
  real<lower=0> sigma_beta;
}
transformed parameters {
  vector[I] alpha = sigma_alpha * alpha_raw;
  vector[J] beta = sigma_beta * beta_raw;
}
model {
  mu ~ normal(0,5);
  alpha_raw ~ normal(0,1);
  beta_raw ~ normal(0,1);
  sigma_alpha ~ normal(0,2);
  sigma_beta ~ normal(0,2);

  for(i in 1:I)
    for(j in 1:J)
      y[i,j] ~ poisson_log(mu + alpha[i] + beta[j]);
}
generated quantities {
  matrix[I,J] lambda;
  for(i in 1:I)
    for(j in 1:J)
      lambda[i,j] = exp(mu + alpha[i] + beta[j]);
}
"

writeLines(stan_code, "model_hmc.stan")
mod <- cmdstan_model("model_hmc.stan")

fit_hmc <- mod$sample(
  data = list(
    I = I,
    J = J,
```

```r
    y = Y
  ),
  chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  seed = 405
)
draws <- fit_hmc$draws("lambda", format = "matrix")

lambda_mean_hmc <- apply(draws, 2, mean)
lambda_lower_hmc <- apply(draws, 2, quantile, 0.025)
lambda_upper_hmc <- apply(draws, 2, quantile, 0.975)

hmc_results <- data.frame(
  mean = lambda_mean_hmc,
  lower = lambda_lower_hmc,
  upper = lambda_upper_hmc
)
```

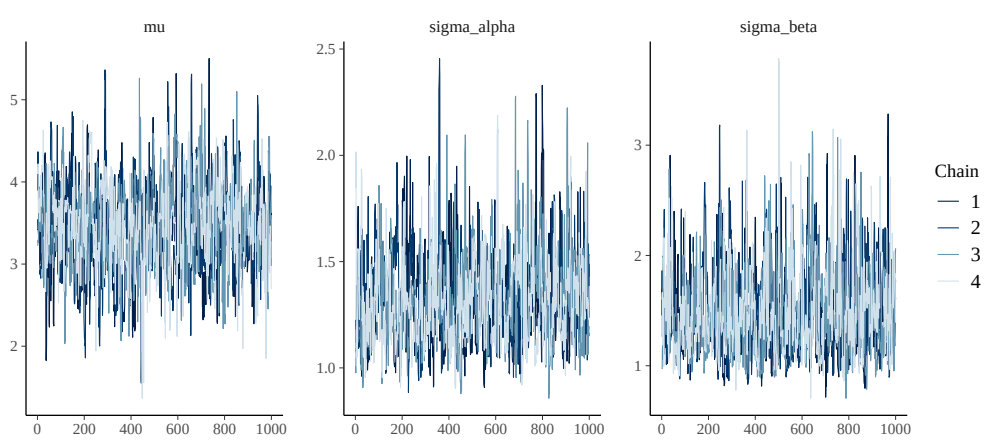
```r
# Summary diagnostics (R-hat and ESS)
summary_hmc <- fit_hmc$summary(c("mu", "sigma_alpha", "sigma_beta"))
print(summary_hmc[, c("variable", "mean", "sd", "rhat", "ess_bulk", "ess_tail")])
```

```
## # A tibble: 3 x 6
##   variable     mean    sd  rhat ess_bulk ess_tail
##   <chr>       <dbl> <dbl> <dbl>    <dbl>    <dbl>
## 1 mu           3.43 0.525  1.01     626.     751.
## 2 sigma_alpha  1.33 0.205  1.01     539.    1004.
## 3 sigma_beta   1.47 0.378  1.00     637.     831.
```

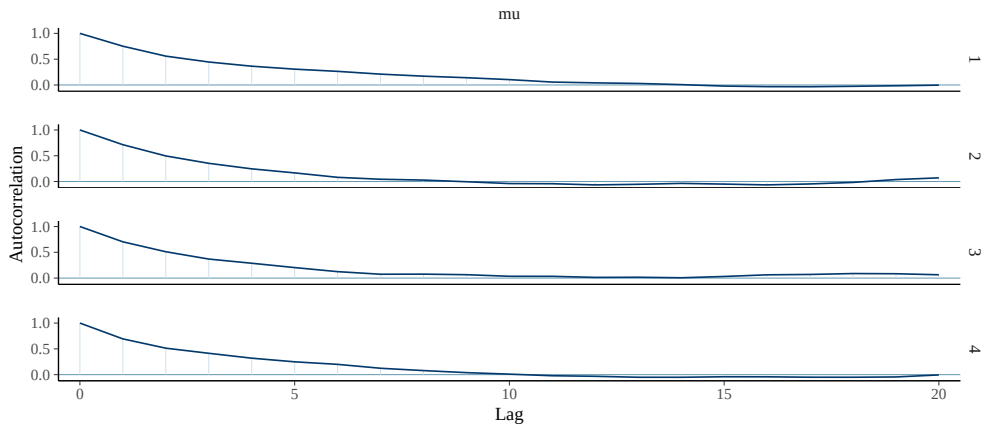
```r
# Trace plots and autocorrelation

draws_array <- fit_hmc$draws(c("mu", "sigma_alpha", "sigma_beta"))

mcmc_trace(draws_array, pars = c("mu", "sigma_alpha", "sigma_beta"))
```



```r
mcmc_acf(draws_array, pars = c("mu"))
```



References

[1] Gelman, A., Carlin, J., Stern, H., Dunson, D., Vehtari, A., & Rubin, D. (2013). *Bayesian Data Analysis* (3rd ed.). CRC Press.